SQL Project: Transportation Planning & Logistics Cost Optimization

Objective:

In this project, you will design and build a relational database for a **Logistics and Transportation** company experiencing **inefficient transportation planning and high logistics costs**.

Transportation data is often scattered across carriers, Transportation Management Systems (TMS), emails, spreadsheets, and manual logs. This makes it difficult for companies to understand transportation performance, optimize routes, control freight costs, and identify recurring delivery problems.

Your goal is to create a centralized SQL database that tracks **shipments, carriers, routes, vehicles, customers, delivery performance, and transportation costs**.

You will create the database, define relationships between tables, populate the database with realistic sample data, and write SQL queries that transform the raw transportation data into useful business insights.

The project is designed to expose you to important SQL concepts including:

- Database and table creation
- Primary and foreign keys
- INSERT, UPDATE, and DELETE
- SELECT statements
- WHERE and filtering
- ORDER BY
- DISTINCT
- Aggregate functions
- GROUP BY and HAVING
- CASE statements
- INNER JOIN and LEFT JOIN
- Subqueries
- Common Table Expressions (CTEs)
- Date and time functions
- String functions
- NULL handling
- Window functions
- Ranking
- Views

- Data validation and constraints
- Business KPI calculations

Problem Statement:

Inefficient Transportation Planning & High Logistics Costs

Transportation information is currently scattered across carriers, TMS systems, emails, spreadsheets, and manual transportation logs.

As a result, management has limited visibility into:

- Freight spending
- Carrier performance
- Route efficiency
- Vehicle utilization
- Late deliveries
- Delivery times
- Cost per shipment
- Cost per kilometer
- Frequently delayed routes
- High-cost transportation lanes

Poor visibility has contributed to inefficient route planning, missed delivery windows, unnecessary transportation expenses, and difficulty evaluating carrier performance.

Management wants a centralized transportation database that can be analyzed using SQL to identify opportunities to **reduce logistics costs and improve delivery performance**.

Company Overview

Company Name: SwiftRoute Logistics Ltd.

Industry: Transportation & Logistics

Company Description:

SwiftRoute Logistics Ltd. provides transportation and delivery services to businesses across multiple regions.

The company manages shipments using both its own vehicles and third-party transportation carriers. Every day, shipments move between warehouses, distribution centers, and customer locations using different transportation routes.

As the company has grown, transportation information has become increasingly fragmented. Shipment information may exist in the company's TMS, carrier reports, spreadsheets, emails, and manual delivery records.

Management wants to build a centralized relational database that provides a single source of truth for transportation operations.

The database will allow management to analyze shipment activity, freight costs, carrier performance, route efficiency, vehicle utilization, and delivery delays.

Instructions

Part 1: Database Design

Design a relational database capable of supporting transportation and logistics analysis.

Your database should contain the following tables.

Customers

Create a table containing customer information.

Suggested fields:

- `customer_id`
- `customer_name`
- `customer_type`
- `city`
- `region`
- `country`

Carriers

Create a table containing information about transportation providers.

Suggested fields:

- `carrier_id`
- `carrier_name`
- `carrier_type`
- `contact_email`
- `rating`
- `active_status`

Carrier types could include:

- Internal Fleet
- Third-Party Logistics Provider
- Courier
- Freight Company

Vehicles

Create a table containing vehicle information.

Suggested fields:

- `vehicle_id`
- `carrier_id`
- `vehicle_type`
- `vehicle_capacity_kg`
- `registration_number`
- `vehicle_status`

Vehicle status could include:

- Available
- In Transit
- Maintenance
- Out of Service

Routes

Create a table containing transportation route information.

Suggested fields:

- `route_id`
- `origin`
- `destination`
- `distance_km`
- `estimated_duration_hours`
- `standard_cost`

Shipments

Create a table containing shipment information.

Suggested fields:

- `shipment_id`
- `customer_id`
- `carrier_id`
- `vehicle_id`
- `route_id`
- `shipment_date`
- `expected_delivery_date`
- `actual_delivery_date`
- `shipment_weight_kg`
- `shipment_status`

Shipment status could include:

- Pending
- In Transit
- Delivered
- Delayed
- Cancelled

Transportation Costs

Create a table containing costs associated with individual shipments.

Suggested fields:

- `cost_id`
- `shipment_id`
- `fuel_cost`
- `driver_cost`
- `toll_cost`
- `carrier_fee`
- `maintenance_cost`
- `other_cost`

Students should calculate the **total transportation cost** from these individual cost components.

Delivery Events

Create a table that records important events occurring during transportation.

Suggested fields:

- `event_id`
- `shipment_id`
- `event_type`
- `event_date`
- `location`
- `notes`

Examples of events include:

- Shipment Picked Up
- Vehicle Breakdown
- Traffic Delay
- Warehouse Delay
- Delivery Attempt
- Shipment Delivered

Part 2: Create the Database

Use SQL `CREATE DATABASE` and `CREATE TABLE` statements to build your transportation database.

Use appropriate SQL data types including:

- INT
- VARCHAR
- DECIMAL
- DATE
- DATETIME
- TEXT
- BOOLEAN

Where appropriate, implement:

- PRIMARY KEY
- FOREIGN KEY
- NOT NULL
- UNIQUE
- DEFAULT
- CHECK constraints

Part 3: Populate the Database

Populate your database using SQL.

Create at least:

- **10 customers**
- **5 carriers**
- **10 vehicles**
- **10 routes**
- **50 shipments**
- **50 transportation cost records**
- **30 delivery event records**

Your sample data should contain enough variation to make the analysis meaningful.

Include examples of:

- Successful deliveries
- Late deliveries
- Cancelled shipments
- Long-distance routes
- Short-distance routes
- High-cost shipments
- Low-cost shipments

- Different carriers
- Different customers
- Different regions
- Vehicle breakdowns
- Traffic delays
- Missing or NULL values where realistically appropriate

Part 4: Basic SQL Data Exploration

Before performing advanced analysis, explore the database.

Query 1: Retrieve Shipment Records

Display all shipments and their basic information.

Query 2: Filter Delayed Shipments

Find shipments with a status of **Delayed**.

Then identify shipments where the actual delivery date was later than the expected delivery date.

Query 3: Explore Unique Transportation Data

Identify unique:

- Shipment statuses
- Carrier types
- Vehicle types
- Regions
- Origins
- Destinations

Part 5: Aggregate Transportation Analysis

Query 4: Shipment Volume by Carrier

Calculate the total number of shipments handled by each carrier.

Display:

- Carrier
- Total shipments

Query 5: Transportation Cost Analysis

Calculate the total cost of every shipment using:

Fuel Cost + Driver Cost + Toll Cost + Carrier Fee + Maintenance Cost + Other Cost

Then calculate:

- Total logistics expenditure
- Average shipment cost
- Minimum shipment cost
- Maximum shipment cost

Query 6: High-Cost Carriers

Calculate total transportation spending for each carrier.

Only return carriers whose total transportation cost exceeds a threshold selected by the student.

Part 7: SQL JOIN Analysis

Query 7: Complete Shipment Report

Create a report containing:

- Shipment ID
- Customer name
- Carrier name
- Vehicle type
- Origin
- Destination
- Shipment date
- Expected delivery date
- Actual delivery date
- Shipment status

This information exists across several tables and must be combined.

Query 8: Customers With and Without Shipments

Return every customer, including customers who have never placed a shipment.

Part 7: Delivery Performance Analysis

Query 9: On-Time Delivery Rate by Carrier

Calculate the percentage of shipments delivered on or before their expected delivery date for each carrier.

The report should contain:

- Carrier name
- Total delivered shipments
- On-time shipments
- Late shipments
- On-time delivery percentage

Query 10: Most Delayed Routes

Identify routes experiencing the largest number of late deliveries.

Display:

- Origin
- Destination
- Total shipments
- Late shipments
- Late delivery percentage

Sort the results from the highest delay rate to the lowest.

Part 8: Subqueries and CTEs

Query 11: Above-Average Transportation Costs

Find shipments whose transportation costs are greater than the average transportation cost across all shipments.

The solution should use a **subquery**.

Query 12: High-Cost Routes Using a CTE

Create a Common Table Expression that calculates transportation spending by route.

Use the CTE to identify routes whose average shipment cost is higher than the overall route average.

Part 9: Window Functions

Query 13: Rank Carriers by Transportation Cost

Rank carriers from highest to lowest based on total transportation expenditure.

Query 14: Rank Routes by Delivery Performance

Rank routes according to their on-time delivery percentage.

Query 15: Running Logistics Cost

Calculate the cumulative transportation expenditure over time.

Part 10: Data Cleaning and NULL Handling

Query 16: Identify Missing Transportation Data

Find records containing missing:

- Delivery dates
- Carrier information
- Vehicle information
- Transportation costs

Part 11: Data Modification

Students should demonstrate their ability to maintain a live database.

Query 17: INSERT

Add a new customer, shipment, and transportation cost record.

Query 18: UPDATE

Update a shipment from **In Transit** to **Delivered** and record its actual delivery date.

Query 19: DELETE

Delete a test record that was entered incorrectly.

Students must explain why `WHERE` clauses are especially important when executing `UPDATE` and `DELETE` statements.

Business Questions

After completing the SQL analysis, answer the following questions using evidence from your database:

1. Which carriers handle the most shipments?
2. Which carriers generate the highest transportation costs?
3. Which carriers have the highest number of delayed shipments?
4. Which routes experience recurring delivery delays?
5. Which routes have unusually high costs per kilometer?
6. Which customers generate the highest logistics spending?
7. What are the most common causes of transportation delays?
8. How has transportation spending changed over time?
9. Are longer routes always more expensive?
10. Where should management investigate opportunities to reduce logistics costs?

Your answers should be based on SQL query results rather than assumptions.

Deliverables

SQL Scripts

Submit SQL scripts containing:

- Database creation
- Table creation
- Primary and foreign keys
- Constraints
- Sample data insertion
- Data exploration queries
- JOIN queries
- Aggregate queries
- Subqueries
- CTEs
- Window functions
- Data modification queries
- SQL views
- Business analysis queries

SQL Analysis Report

Submit a short report explaining:

Database Design

Explain why each table was created and how the tables relate to each other.

Query Explanation

For the major analytical queries, explain:

- What business question the query answers
- What SQL concepts were used
- How the query works
- What the results mean

Sample Output

Provide screenshots or query outputs for at least **five major analytical queries**.

Business Recommendations

Based on your SQL analysis, provide at least **five findings or recommendations** that could help SwiftRoute Logistics Ltd. improve transportation planning, reduce logistics costs, or improve delivery performance.

Recommendations must be supported by results from your SQL queries.

Resources & Timeline:

Submission Deadline: Oct 15th, 2026 - 7:00 PM WAT
Review Presentation: Oct 17th, 2026 - 8:00 PM WAT